

Distinction, Reentry, and the Minimal Numeric Ego

Abstract

This essay develops a minimal model of ego formation from distinction, marking, and reentry. The philosophical claim is that structure, meaning, and selfhood need not be grounded in external semantic reference, but can arise from recursive constraint propagation within a randomly seeded process. A simple two-bit numerical model illustrates this claim: despite stochastic input and unbiased marginal statistics, deterministic update rules induce nonzero mutual information between algorithm choice and output. The resulting structure suggests a distinction between ego, understood as representable recursive memory, and self, understood as the unrepresentable source of constraint that conditions the evolution of that memory.

1 Philosophical Motivation: Distinction, Marking, and Reentry

Any account of a describable universe must implicitly assume the possibility of differentiation. Prior to objects, properties, or truth claims, there must exist the capacity to separate one state from another. This primitive condition is expressed as the drawing of a *distinction*. A distinction represents the minimal form of intelligibility, establishing the separation between alternatives such as this and not-this, true and false, or inside and outside.

By itself, however, a distinction does not yet produce a world. A world arises only when one side of the distinction is *marked*, thereby breaking the symmetry between possibilities and introducing orientation. Marking is the primordial generative act underlying the emergence of structure.

Once a mark has been made, a further operation becomes possible: *reentry*. Reentry denotes the participation of prior distinctions within subsequent ones. The system recursively refers to its own products, so that each new act of marking is conditioned by the accumulated consequences of earlier acts. Prior marks function as implicit constraints, effectively serving as axioms that regulate future operations.

The initial mark is necessarily arbitrary. In the absence of prior structure capable of constraining orientation, the first selection cannot be guided by information internal to the system. It is therefore natural to model this act as random. Likewise, the appearance of distinctions themselves need not be grounded in external necessity and may be treated as stochastic events. (In general, the space of possible distinctions need not be limited; many different forms of differentiation may arise. The particular restriction to a finite set of distinguishable types will be introduced only when constructing a minimal computational model.) From these premises, the recursive process unfolds: distinctions arise, marks are chosen, and the results of these choices become conditions for subsequent choices.

Within this perspective, what is ordinarily described as the ego corresponds to the stabilized record of prior marking decisions. Its content is not an independent entity, but a recursively reproduced constraint structure. It may be represented formally as a growing bitstream, where each bit encodes a marking decision and each decision contributes to the conditions governing future decisions.

Yet the bitstream itself carries no intrinsic semantic content. No individual bit refers beyond the distinction system that generates it. Apparent meaning arises not from representational correspondence but from relational structure within the recursive process.

Because both the initial mark and the sequence of distinctions are modeled as random, no absolute grounding of meaning is available. Nonetheless, structure and regularity emerge. Meaning, in this formulation, is not defined as reference to an external reality but as correlation between the bitstream and the fluctuation of distinction types within the evolving system. Lawfulness becomes an internal property of the process rather than an imposed feature of an independently existing domain.

The predicament of being may therefore be understood as the consequence of treating the resolution of distinctions as necessary rather than optional. Once marking is taken as compulsory, the recursive machinery of coherence, memory, and identity arises. Prior marks constrain subsequent ones, generating the appearance of lawfulness, continuity, and selfhood.

2 A Minimal Numeric Model of Ego

In the general framework described above, the space of possible distinctions and the length of the ego bitstream are unrestricted. To construct a tractable numerical system simulating the model, we seek the simplest possible universe that still captures the salient feature of emergent coupling between the world and ego memory. In a pathological world with only one kind of distinction, only one bit of memory is required for the optimal solution: the distinction will be marked in the same way forever once the first mark is made. Little structure can arise in such a system.

We therefore consider a marginally richer case: at each step, one of two possible kinds of distinction may appear. This restriction allows the input to be represented by a single stochastic variable $A_t \in \{1, 2\}$, selecting between two deterministic update rules. These rules act on a 2-bit ego memory (b_{t-1}, b_t) and produce a binary marking outcome b_{t+1} for the current distinction. The resulting bit is appended to the ego bitstream and becomes part of the history used in the next step.

The update rules are 2-input/1-output Boolean functions. Among the 120 possible unique pairs of such functions (see Appendix B), we restrict attention to those that generate unbiased output, are invariant under global bit-flip, and exhibit nontrivial but nondegenerate coupling between input and output. These constraints significantly reduce the space of admissible pairs. Among the simplest representatives satisfying these properties are the pairs $[\text{NOR}(x, y)/\text{NAND}(x, y)]$ and $[(\neg x) \wedge y / (\neg x) \vee y]$, which are related by global inversion and yield identical statistical behavior. In what follows, we work with the latter for convenience.

2.1 Formal description

Let the internal state at time t be $(x, y) = (b_{t-1}, b_t)$, and let the algorithm choice $A_t \in \{A_1, A_2\}$ be selected uniformly at random. Define $A_1(x, y) = (\neg x) \wedge y$ and $A_2(x, y) = (\neg x) \vee y$. The next output bit is $b_{t+1} = A_t(x, y)$, and the next internal state is (y, b_{t+1}) .

The algorithm choice is stochastic, and the output bitstream is marginally unbiased. Nevertheless, the deterministic constraints embedded in the update rules propagate dependencies through the joint process. These dependencies may be quantified using the mutual information between the algorithm choice A_t and the produced bit b_{t+1} :

$$I(A; b) = \sum_{a,b} P(a, b) \log_2 \frac{P(a, b)}{P(a)P(b)},$$

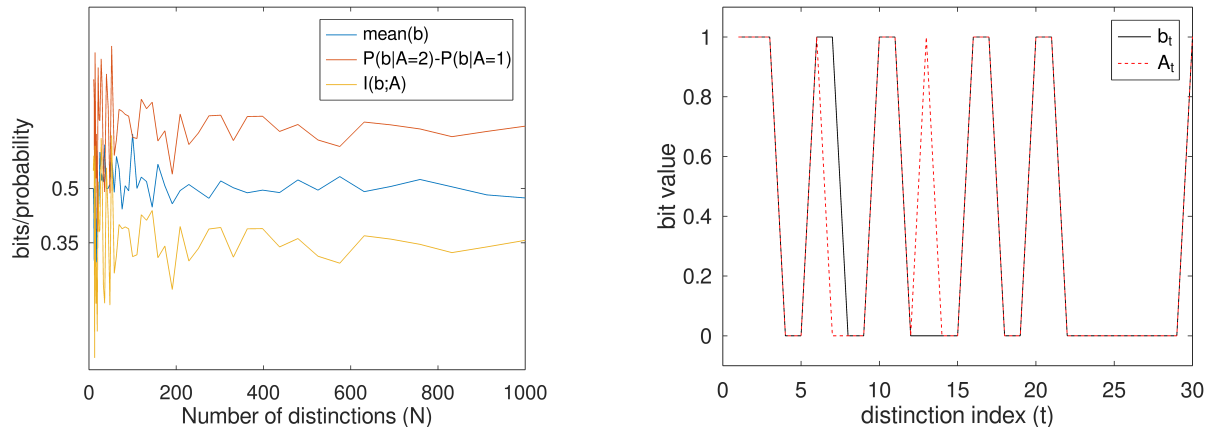
where $a \in \{1, 2\}$.

The central point is that structure appears not in either sequence A_t or b_t considered in isolation, but in their joint organization. Meaning, in this minimal sense, is mutual information generated by recursive constraint imposed by the logic rules.

2.2 Empirical results

The behavior of the model described above can be examined numerically using the GNU-Octave / Matlab implementation provided in Appendix A. Figure 1 shows representative results obtained from this simulation. The left panel displays ensemble statistics computed from independent runs of increasing length, demonstrating convergence of the output mean, conditional separation, and mutual information. Despite the stochastic selection of algorithms and the absence of bias in the output bitstream, the mutual information $I(A; b)$ approaches a stable nonzero value of approximately 0.35 bits.

The right panel shows a single realization of the process, illustrating how this dependence manifests locally in the coupling between the stochastic input A_t and the generated bitstream $b(t)$. These numerical observations provide empirical evidence that structured statistical dependence emerges from the recursive constraints of the update rules alone. In the following section, this behavior is derived analytically using a four-state Markov model, yielding the same value for the mutual information.



(a) Global convergence of mean output, conditional separation, and mutual information.

(b) Example timestep behavior illustrating the emergence of statistical dependence between stochastic input A_t and output b_t .

Figure 1: Emergent structure in the minimal ego model. Despite stochastic inputs and unbiased marginals, nonzero mutual information arises from recursive constraint imposed by the logical algorithm set. The left panel shows global convergence, while the right panel illustrates the local mechanism by which dependence is generated.

3 Markov Analysis

The numerical results above suggest the emergence of a stable, nonzero mutual information between the stochastic input A_t and the generated bitstream. To understand the origin of this value and derive it analytically, we pass to a Markov description of the system in which the 2-bit ego

memory (b_{t-1}, b_t) defines a four-state process with deterministic rule-conditioned transitions. This reformulation makes explicit the structure implicit in the recursive update, at the cost of enlarging the state space.

Label the four states as

$$00, \quad 01, \quad 10, \quad 11.$$

The induced transition matrix is

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}.$$

Solving

$$\pi = \pi P$$

gives the stationary distribution

$$\pi(00) = \frac{1}{3}, \quad \pi(01) = \frac{1}{6}, \quad \pi(10) = \frac{1}{6}, \quad \pi(11) = \frac{1}{3}.$$

(Notice that $P(b_{t+1} = 1) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$, confirming the output bitstream is unbiased.)
Now define

$$p_1 = P(b_{t+1} = 1 \mid A_1),$$

and

$$p_2 = P(b_{t+1} = 1 \mid A_2).$$

Under A_1 , output is 1 only in state 01, so

$$p_1 = \frac{1}{6}.$$

Under A_2 , output is 0 only in state 10, so

$$p_2 = \frac{5}{6}.$$

Hence the conditional separation is

$$C = p_2 - p_1 = \frac{2}{3}.$$

The joint distribution is

	$b = 0$	$b = 1$
A_1	5/12	1/12
A_2	1/12	5/12

with marginals

$$P(A_1) = P(A_2) = \frac{1}{2}, \quad P(b = 0) = P(b = 1) = \frac{1}{2}.$$

Therefore

$$I(A; b) = \frac{5}{12} \log_2 \left(\frac{5/12}{1/4} \right) + \frac{1}{12} \log_2 \left(\frac{1/12}{1/4} \right) + \frac{1}{12} \log_2 \left(\frac{1/12}{1/4} \right) + \frac{5}{12} \log_2 \left(\frac{5/12}{1/4} \right).$$

Equivalently,

$$I(A; b) = \frac{5}{6} \log_2 \left(\frac{5}{3} \right) + \frac{1}{6} \log_2 \left(\frac{1}{3} \right) \approx 0.3499776 \text{ bits.}$$

The system therefore has

$$\text{mean}(b) = 0.5, \quad p_1 = \frac{1}{6}, \quad p_2 = \frac{5}{6}, \quad I(A; b) \approx 0.35 \text{ bits.}$$

The Markov formulation provides a complete analytic account of the observed mutual information by making explicit the transition structure over the four possible internal states. However, this description raises a natural question: does the full 2-bit state (b_{t-1}, b_t) represent the minimal structure required to capture the dependence between A_t and b_{t+1} , or does it contain redundant information?

The mutual information depends only on the joint distribution $P(A, b)$, not on the full internal state itself. It is therefore natural to ask whether a reduced description of the state exists that preserves this dependence while eliminating unnecessary detail. This leads to the problem of identifying a minimal sufficient statistic of the internal state with respect to the variable A_t .

4 Compression and Minimal Sufficient Statistic

We now show that the full 2-bit internal state can be reduced to a single effective bit without loss of mutual information. The relevant quantity is not the full state (b_{t-1}, b_t) , but only whether the two bits are equal or different.

The full observer state is the two-bit memory

$$(x, y) = (b_{t-1}, b_t).$$

However, the two algorithms differ only in those states where the two bits are equal, and agree when they are not. Thus the full state contains more detail than is necessary. The relevant distinction is simply whether the two bits are the same or different.

Define the compressed variable

$$S = \text{XNOR}(x, y),$$

so that

$$S = \begin{cases} 1, & x = y, \\ 0, & x \neq y. \end{cases}$$

Thus

$$00 \mapsto 1, \quad 01 \mapsto 0, \quad 10 \mapsto 0, \quad 11 \mapsto 1.$$

The observer does not need to remember both bits. It only needs the distinction: same versus different. This compressed one-bit statistic preserves the same mutual information between algorithm choice and observed output.

The ego is therefore not the full memory of the world, but the minimal distinction that makes the world informative.

4.1 Closure, Grounding, and the Emergence of a Minimal Observer

The appearance of the compressed variable

$$S = \text{XNOR}(b_{t-1}, b_t)$$

reveals a structural tension between closure and grounding. Although the full two-bit state (b_{t-1}, b_t) provides a complete description of the system at a given level, it does not, by itself, identify which aspects of that state are relevant to the generative process. From the standpoint of the closed four-state ontology, all states are equally admissible, and no preferred coarse-graining is singled out.

The reduction to S is therefore not a property of the state space considered in isolation. Rather, it arises only when the state is viewed in relation to the update rules A_1 and A_2 , or equivalently, in relation to the statistical coupling between algorithm choice and output. The variable S captures precisely those distinctions that remain informative about the rule-conditioned dynamics. It is thus defined not intrinsically, but relationally.

This demonstrates that a closed description of state is insufficient to ground the notion of relevance. The full two-bit ontology is complete in the sense of closure, but it does not determine which degrees of freedom carry information about the process that generates it. Conversely, identifying the minimal sufficient statistic requires reference to structure that is not contained within the instantaneous state alone—namely, the transition rules or the induced joint distribution.

In this sense, the system contains its own minimal description, but cannot select it from within its closed representation. The emergence of S therefore suggests a more general structural feature: the identification of relevant distinctions depends on relations that are not fully specified at the level of the states themselves.

This observation will be developed further in the following section, where the relationship between representable structure and the source of constraint that organizes it is examined more explicitly.

5 Interpretation and Consequences

This construction reveals a basic structural fact: meaningful statistical dependence can arise purely from internal constraints acting on stochastic input. No reference to an external world model or symbolic representation is required. The system does not understand anything in the ordinary semantic sense, yet it produces a structured coupling between its internal state and its input.

A crucial feature of the model is that the rules governing state evolution are not representable within the system’s instantaneous state. At any given moment, only the current bits are present. The law that transforms them is not itself encoded as one of those bits.

One can expand the ontology by passing to a Markov representation in which four states encode the possibilities of the two-bit memory. But this merely internalizes previously external structure at the cost of increasing descriptive complexity. At the new level, a new form of law again appears that is not fully representable within that ontology.

Thus no fixed level of description can be both closed and fully self-grounding.

This suggests a minimal distinction between ego and self. The ego is the observable, representable structure: the evolving bitstream together with its induced correlations and memory. It is the part of the system that can be described within the ontology at a given level.

The self, by contrast, corresponds to the source of constraint that shapes this evolution but is not itself an object within the system's representational frame. It is not directly accessible as a state, but only inferred through its effects on the dynamics of the ego.

From this perspective, what might be called experience is not a property of isolated states, but of the ongoing coupling between internal dynamics and uncontrolled external input. The system's internal state does not mirror the world. Nor does it predict it in a strong sense when the input is stochastic. Instead, it maintains a persistent statistical dependence on the input, induced by its own constraints.

This also reframes simulation theory. The sense of living in a simulation need not arise from an external machine or higher-order civilization running a program. Instead, it may emerge from the internal dynamics of distinction, marking, and reentry: the operations through which a system brings forth both a self and a world. The simulation is not imposed from outside; it is the ongoing recursive process by which the system organizes its own experience.

Likewise, the model offers a structural alternative to Nagel's formulation of consciousness as requiring an additional fact: that there is something it is like to be the system. Here, what-it-is-likeness is not treated as a separate property. Appearance is identified with the system's operational capacity to register and stabilize distinctions under recursive constraint. Phenomenology is reinterpreted as the internal organization of distinction-handling processes rather than as an intrinsic qualitative residue.

The model therefore supports a minimal structural claim: any system that generates nontrivial internal structure from stochastic input must rely on constraints that are not fully representable within its current ontology. Identifying this unrepresentable source of constraint with "I" yields a stripped-down notion of self grounded not in representation or meaning, but in the unavoidable boundary between what is expressed and what makes expression possible.

In Wittgenstein's terms, the subject does not belong to the world; it is a limit of the world. In Spencer-Brown's terms, the boundary created by distinction cannot be fully contained within the domain it defines. Each attempt to reenter the boundary shifts the level of description but does not abolish the boundary.

The self is therefore not a fixed entity, but a structural feature of any recursively constrained process: the always-present, never-fully-representable condition of its own evolution.

Appendix

A GNU-Octave / MATLAB Code

```
function ego_minimal()
    close all

    % Octave-compatible seeding (optional)
    rand("seed", sum(100*clock));

    % --- number of distinctions ---
    Ns = round(logspace(1, 3, 51));
```

```

% --- throw away transient for analysis ---
burn_frac = 0.2;

for k = 1:numel(Ns)

    N = Ns(k);
    burn = floor(burn_frac * N);

    % --- random initial 2-bit state ---
    b_prev = (rand < 0.5);
    b_curr = (rand < 0.5);

    % --- initialize algorithm choice record ---
    As = zeros(1, N);

    % --- initialize ego bitstream record ---
    bs = false(1, N);

    % --- iterate through N random distinctions ---
    for t = 1:N

        % --- choose algorithm randomly ---
        if rand < 0.5
            A = 1;
            out = A1(b_prev, b_curr);
        else
            A = 2;
            out = A2(b_prev, b_curr);
        end

        As(t) = A;
        bs(t) = out;

        % --- shift to retain last two ego bits ---
        b_prev = b_curr;
        b_curr = out;

    end

    % --- remove transient ---
    As2 = As(burn+1:end);
    b2 = bs(burn+1:end);

    % --- correlation analysis ---
    p1 = mean(b2(As2==1));
    p2 = mean(b2(As2==2));

    C(k) = p2 - p1;
    I(k) = mi_A_B(As2, b2);
    mb2(k) = mean(b2);

    fprintf('N=%d mean(b)=%.4f p1=%.4f p2=%.4f C=%.4f I=%.4f bits\n', ...
        N, mb2(k), p1, p2, C(k), I(k));

```



```

end

fprintf("\nAnalytic targets:\n");
fprintf('mean(b)=0.5 p1=%.6f p2=%.6f C=%.6f I=0.34998 bits\n', ...
    1/6, 5/6, 2/3);

figure(1)
plot(Ns, mb2, Ns, abs(C), Ns, I)
legend('mean(b)', 'P(b|A=2)-P(b|A=1)', 'MI(b,A)')
set(gca,'fontsize',14,'ytick',[0.34998 0.5], 'xlabel', 'Number of distinctions (N)', '
    ylabel', 'bits/probability')

figure(2)
ptail = 30;
plot(1:(ptail+1), bs((end-ptail):end), 'k', ...
    1:(ptail+1), As((end-ptail):end)-1, 'r--')
legend('b_t', 'A_t')
set(gca,'fontsize',14,'ylim',[-0.1 1.1], 'xlim', [0 ptail], 'xlabel', 'distinction
    index (t)', 'ylabel', 'bit value')

end

function out = A1(x, y)
    out = (~x) & y;
end

function out = A2(x, y)
    out = (~x) | y;
end

function I = mi_A_B(A, b)

    C = zeros(2,2);

    for t = 1:numel(A)
        C(A(t), b(t)+1) = C(A(t), b(t)+1) + 1;
    end

    Pab = C / sum(C(:));
    Pa = sum(Pab,2);
    Pb = sum(Pab,1);

    I = 0;

    for a = 1:2
        for bb = 1:2
            if Pab(a,bb) > 0
                I = I + Pab(a,bb) * log2(Pab(a,bb) / (Pa(a)*Pb(bb)));
            end
        end
    end
end

```

```

end
end

```

B Anticipating an objection

A natural objection to the interpretation of nonzero mutual information in the minimal model is the following. The two update rules,

$$A_1(x, y) = (\neg x) \wedge y, \quad A_2(x, y) = (\neg x) \vee y,$$

have different output statistics when evaluated on independent random inputs. If x and y are independent Bernoulli(1/2) variables, then

$$P(b = 1 \mid A_1) = \frac{1}{4}, \quad P(b = 1 \mid A_2) = \frac{3}{4}.$$

Thus, knowledge of the output bit b provides partial information about which algorithm was applied. From this perspective, the observed mutual information between algorithm choice A_t and output b_{t+1} **might** appear to arise trivially from the differing biases of the two logical gates, rather than from any emergent structure.

This objection would be valid if the inputs to the update rules were independent random variables at each timestep. In that case, the system would reduce to a memoryless channel in which information about A_t is directly encoded in the output distribution, and no persistent structure would arise.

However, this is not the situation in the model. The inputs to the update rules are not independent random bits, but the system's own internal state (b_{t-1}, b_t) , which evolves under the influence of the same rules.

The process therefore forms a closed, recursively constrained dynamical system rather than a sequence of independent evaluations. As a consequence, the distribution over input states is not uniform; at the same time, the marginal output remains unbiased.

The key point is that these conditional probabilities are not intrinsic properties of the gates alone, but arise from the interaction between the gates and the evolving state distribution. The system does not receive independent random inputs; rather, it generates its own inputs through recursive application of its update rules. The observed mutual information therefore reflects structure induced by constraint propagation over time, not direct encoding of algorithm identity into the output.

This can be further clarified by comparison with a modified system in which the inputs (x, y) are replaced at each timestep by independent random bits. In that case, the output probabilities revert to 1/4 and 3/4, and the mutual information drops to $\frac{3}{4} \log_2 \frac{3}{2} + \frac{1}{4} \log_2 \frac{1}{2} \approx 0.19$ bits. More importantly, the process becomes memoryless, and no persistent internal structure or coupling between past and future states is maintained.

The distinction is therefore between two qualitatively different mechanisms:

- In a memoryless system with independent inputs, mutual information arises from direct statistical bias in the output of different rules.
- In the recursive system studied here, mutual information arises from the sustained coupling between stochastic inputs and internally generated state, mediated by deterministic constraints.

Only the latter exhibits the emergence of structured dependence that motivates the interpretation of the bitstream as a minimal “ego” and the rule set as a source of constraint not representable within the state itself.

C Search-Space Explosion

Suppose an algorithm takes n input bits and produces m output bits. There are 2^n possible input states. For each input state, the algorithm may output any one of 2^m possible output states. Therefore the number of distinct algorithms is

$$N_{\text{alg}} = (2^m)^{2^n} = 2^{m2^n}.$$

If we want to choose between two algorithms, the number of ordered pairs is approximately

$$N_{\text{pairs}} \sim 2^{2m2^n}.$$

For the original one-bit world, the algorithm has

$$n = 2, \quad m = 1,$$

so

$$N_{\text{alg}} = 2^{1 \cdot 2^2} = 2^4 = 16.$$

The number of unordered non-identical pairs is

$$\binom{16}{2} = 120.$$

For a two-bit world with one timestep of memory, the algorithm has

$$n = 4, \quad m = 2.$$

Therefore

$$N_{\text{alg}} = 2^{2 \cdot 2^4} = 2^{32},$$

and the number of ordered pairs is approximately

$$N_{\text{pairs}} \approx (2^{32})^2 = 2^{64} \approx 10^{19}.$$

This is the combinatorial explosion. Adding one more world bit does not merely double the search space. It doubles the number of input states, and each input state must be assigned an m -bit output.

C.1 Multi-bit world, with bit-noise

The multi-bit extension introduces a new regime that is not present in the minimal two-bit model, when memory must be maintained across many degrees of freedom in the presence of noise. In such systems, independent storage (parallel bitstream channels) of each bit leads to an exponential decay of recoverable information over time. By contrast, relational encoding — where present degrees of freedom $\mathbf{b}_t$ are constrained by consistency relations spanning multiple bits — admits a finite error

threshold, allowing information to persist despite ongoing perturbations. This distinction does not depend on the specific mechanism generating the input distinctions; it follows from the structural tradeoff between capacity and robustness in any finite system.

This suggests that the emergence of correlated phases beyond an error threshold is not an added assumption but a generic feature of the multi-bit world, with a thermodynamic preference to preserve information that is streaming out of finite memory. When the cost of maintaining independent memory becomes too high, the system is forced to reallocate its degrees of freedom toward enforcing cross-bit constraints. In this regime, information is no longer stored locally in individual markings, but in the relational structure that binds them. Thus, the requirement of consistent marking over time naturally leads, in larger systems, to the formation of distributed constraint structures that preserve distinctions collectively rather than independently.

D Closing Note

The *Hsin Hsin Ming* expresses the same structural intuition in religious language:

The Great Way is not difficult for those who have no preferences. When love and hate are both absent, everything becomes clear and undisguised. Make the smallest distinction, however, and heaven and earth are set infinitely apart.